

Pedagogically Grounded AI Cognition

A developmental hybrid architecture for structurally modifiable machine reasoning

Concept paper and technical design synthesis

Core thesis: AI cognition should be treated as a structurally modifiable, developmentally staged, pedagogically mediated process in which learned models remain useful, but semantic authority is progressively transferred to explicit schemas, guided reasoning, working memory, and verification.

ABSTRACT

This paper consolidates a hybrid view of artificial intelligence in which education is not merely an application domain, but a theory of how cognition itself should be formed. The proposal draws on Feuerstein's Structural Cognitive Modifiability, Piagetian developmental staging, Vygotskian mediation and scaffolded learning, curriculum learning in machine learning, symbol emergence, and contemporary neuro-symbolic research. Its central claim is that large learned models may remain useful for fluency, retrieval, proposal generation, and perception, but should not remain sovereign over meaning. Instead, semantic claims should be governed by explicit schema memory, a mediation controller, stage-aware development, an inspectable reasoning workspace, and a verification layer that can refuse, qualify, or hand back. The result is a conception of AI as a governed developmental mind: one that can be taught, corrected, and restructured over time rather than treated as a fixed statistical competence that merely requires patching. The second half of the paper translates this educational theory into a practical systems design with identifiable integration points for hybrid implementation.

1. Framing the problem

The dominant narrative in contemporary AI still treats reasoning primarily as an achievement of large-scale pattern learning. On this view, a sufficiently capable model learns broad regularities from immense corpora and then receives auxiliary additions such as retrieval, tools, memory, or constitutional constraints. This approach has been extraordinarily productive, but it also inherits familiar weaknesses: overconfident synthesis, brittle grounding, shallow transfer in edge cases, and difficulty separating fluency from earned understanding.

The present proposal does not reject machine learning. It rejects the assumption that statistical competence should remain the sole or highest court of semantic authority. The alternative is a hybrid architecture in which learned models remain productive at the substrate level, while educational-developmental structures govern how understanding is formed, revised, and released. In that sense, the question shifts from 'How do we make a model more capable?' to 'What laws should govern the formation of capability itself?'

2. Educational foundations

2.1 Structural Cognitive Modifiability

Feuerstein's theory begins with a deeper premise than ordinary performance optimization: intelligence is not a fixed quantity, but a process that can be structurally modified through mediated experience and the right environment. Applied to AI, this suggests that failure should not be treated only as wrong output. It may indicate the need for reorganization of schemas, revised mediation, or a change in developmental pathway.

2.2 Developmental staging

Piaget contributes the principle that cognition matures through stages rather than arriving in one act of total fluency. A machine system may therefore need progression gates: simple contingencies before robust relations, concrete task reasoning before abstraction, and abstraction before metacognitive self-revision. A fluent answer should not automatically count as mature understanding.

2.3 Mediation and the Zone of Proximal Development

Vygotskian theory shifts emphasis from isolated inference to guided formation. In an AI setting, mediation is more than a user-facing teaching style. It becomes a control function that decides whether to answer directly, ask for clarification, decompose the task, offer a scaffold, or explicitly hand back because the problem exceeds current warranted competence.

2.4 Constructed meaning

Work on symbol emergence and developmental robotics further suggests that meaning is not simply assigned to inert tokens. It is constructed through recurrent interaction, social communication, and multimodal contact with the world. This supports the idea that semantics should be stabilized through revisable schemas and grounding procedures rather than resting entirely inside pretrained weights.

Educational foundation	Core claim	Architectural implication
Feuerstein	Cognition is structurally modifiable, not fixed.	Introduce restructuring loops, revision histories, and mediated correction.
Piaget	Competence develops in stages.	Use developmental gates, stage-aware curricula, and progression criteria.
Vygotsky	Learning is strengthened through guided mediation.	Add a controller for scaffolding, questioning, and calibrated support.
Symbol emergence	Meaning is built through interaction and grounding.	Externalize schemas and connect them to evidence, context, and world contact.

3. Principles of a pedagogically grounded hybrid

- **Learned models may propose, but may not rule.** Neural systems remain useful for linguistic fluency, retrieval cues, multimodal compression, and generative proposals. However, they should not have final authority over warranted claims.
- **Schemas become the semantic spine.** Concepts should be represented as explicit, revisable structures containing relations, preconditions, expected outcomes, constraints, evidence links, and revision history.
- **Memory must preserve formative history.** A developmental system requires episodic memory of encounters, pedagogical memory of what kinds of mediation worked, and constitutional memory of non-negotiable norms.
- **Reasoning must occur in an inspectable workspace.** Task decomposition, uncertainty tracking, contradiction checks, and schema composition should occur in a visible reasoning layer rather than disappearing entirely into opaque activations.
- **Handback is a mark of mature cognition.** The system should be permitted to withhold, qualify, redirect, or refuse when it has not earned a stable answer.

4. Technical design

The educational theory can be implemented as a layered hybrid stack. The design below preserves a learned substrate, but wraps it inside memory, mediation, developmental control, an explicit reasoning workspace, and final adjudication.

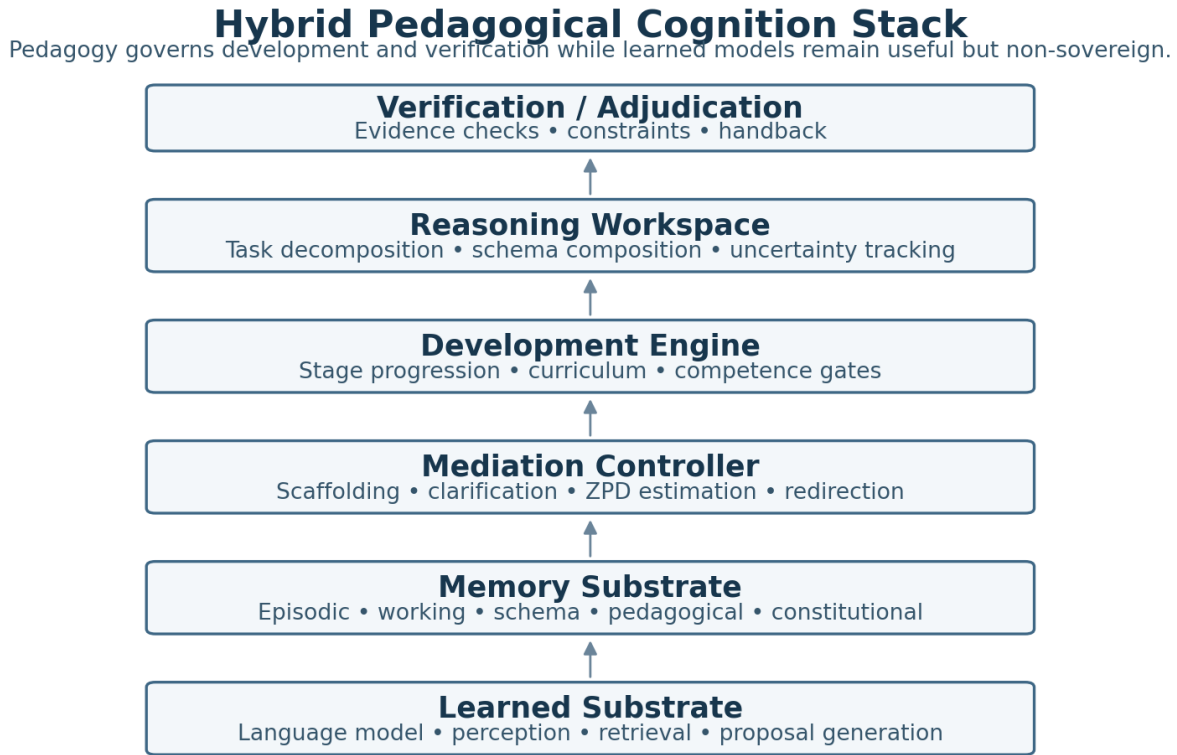


Figure 1. A layered hybrid stack in which pedagogy governs development and verification while learned models remain productive but non-sovereign.

4.1 Learned substrate

A compact or moderate learned model handles language generation, perceptual encoding, retrieval matching, and proposal generation. Its outputs are candidates rather than verdicts.

4.2 Memory substrate

This layer contains episodic memory, working memory, schema memory, pedagogical memory, and constitutional memory. Together they preserve both content and the history of how content was formed or corrected.

4.3 Mediation controller

This module estimates whether the system should answer directly, request clarification, decompose the task, offer a hint, redirect, or hand back. It operationalizes scaffolded learning and calibrates support to current competence.

4.4 Development engine

This engine tracks what the system has genuinely mastered. It decides when a schema is provisional, stable, contradicted, or ready to support higher-order abstraction. It also governs curriculum progression and competence thresholds.

4.5 Reasoning workspace

This is the active cognitive field in which relevant schemas are assembled, task steps are planned, uncertainty is tracked, contradictions are compared, and interim judgments are held for review.

4.6 Verification and adjudication

Before release, a candidate answer passes evidence checks, schema consistency checks, developmental appropriateness checks, and constitutional constraints. This is the layer that can qualify, refuse, or escalate.

4.7 Restructuring loop

When failure occurs, the system asks whether the problem lay in missing schemas, malformed schemas, weak mediation, poor retrieval, stage mismatch, or overconfident release. The target is structural revision, not merely output repair.

5. Integration points for implementation

A practical prototype should begin as an overlay on an existing smaller language model rather than as a fully novel cognition stack. The following integration points offer a realistic path from concept to system.

Schema store: Define an external schema object with fields for concept type, relations, constraints, evidentiary links, contradiction markers, confidence, developmental level, and revision history.

Working-memory trace: Create an explicit task object that records decomposition, active schemas, competing hypotheses, unresolved tensions, and release conditions.

Mediation policy: Add a control policy that can choose among direct answer, clarifying question, scaffolded hint, decomposition, retrieval, refusal, or handback.

Stage gating: Define stage-level criteria such as recognition, relation formation, procedural use, explanation, transfer, and self-correction. A capability is only released as owned once it passes its gate.

Verification pipeline: Require evidence grounding, consistency checks, constraint checks, and calibrated uncertainty before an answer becomes final.

Encounter logging: Store each meaningful interaction as an episode, including what failed, what intervention helped, and whether the system had to revise a schema.

Evaluation: Measure not only task accuracy, but transfer, calibration, reduction of hallucination, improved handback, and robustness of self-correction over time.

6. Research agenda

- How should a schema be formally represented so that it is both machine-usable and semantically inspectable?
- What criteria distinguish assimilation from accommodation in a machine developmental setting?
- How can developmental readiness be measured without collapsing back into surface benchmark scores?
- What kinds of mediation improve genuine transfer rather than merely local task success?
- How much of semantic authority can be externalized into explicit structures before fluency degrades unacceptably?
- What forms of world contact or simulation are sufficient to ground concepts beyond text-only regularities?

7. Closing synthesis

The proposal developed here is neither a rejection of contemporary AI nor a nostalgic return to brittle symbolic systems. It is a hybrid claim: learned models remain valuable, but pedagogy should become part of the architecture of cognition itself. A system designed in this way would be taught, corrected, staged, and restructured rather than merely scaled. Its greatest difference from mainstream practice would not be the absence of learning, but a change in what is allowed to count as understanding.

References

- Delvecchio, G. P., et al. (2026). Neuro-Symbolic Artificial Intelligence: A Task-Directed Systematic Review. arXiv:2603.03177.
- Feuerstein, R. (1990/2023 reprint). The Theory of Structural Cognitive Modifiability and Mediated Learning Experience. Feuerstein Institute.
- Patania, S., Annese, L., Koyuturk, C., Ruggeri, A., & Ognibene, D. (2025). Dialogic Social Learning for Artificial Agents: Enhancing LLM Ontology Acquisition through Mixed-Initiative Educational Interactions. arXiv:2507.21065.
- Taniguchi, T., Nagai, T., Nakamura, T., Iwahashi, N., Ogata, T., & Asoh, H. (2016). Symbol Emergence in Robotics: A Survey. *Advanced Robotics*, 30(11-12), 706-728.
- Wang, X., Chen, Y., & Zhu, W. (2020). A Survey on Curriculum Learning. arXiv:2010.13166.